

Lingxi Zhang

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Research Interests : Multi-Agent Systems, Knowledge Engineering, Large Language Models, NLP

EDUCATION

Rice University

Houston, TX

- Ph.D. in Computer Science

08 / 2024 – Now

Renmin University of China

Beijing, China

- M.Sc. in Computer Application Technology

09 / 2021 – 06 / 2024

Renmin University of China

Beijing, China

- B.E. in Computer Science and Technology
- B.Sc. in Mathematics and Applied Mathematics | Minor

09 / 2017 – 06 / 2021

PUBLICATION

- [1] Jiamu Zhang*, **Lingxi Zhang***, Pengjun Lu*, et al. Rethinking the Evaluation of Efficiency Methods for Multi-Agent Systems. In *EMNLP'26*, Oct, 2026
- [2] **Lingxi Zhang**, Guangtao Zheng and Hanjie Chen. When Embedding-Based Defenses Fail: Rethinking Safety in LLM-Based Multi-Agent Systems. In *ICML'26*, Jul, 2026
- [3] **Lingxi Zhang**, Yu-Neng Chuang, Guanchu Wang, Ruixiang Tang, Xuanning Cai, Rajesh Shenoy and Xia Hu. A Decoupled Multi-Agent Framework for Complex Text Style Transfer. In *EMNLP'25 findings*, 2025.
- [4] **Lingxi Zhang**, Yue Yu, Kuan Wang and Chao Zhang. ARL2: Aligning Retrievers for Black-box Large Language Models via Self-guided Adaptive Relevance Labeling. In *ACL'24*, pages 3708–3719, Aug. 2024
- [5] Yanling Wang, Jing Zhang, **Lingxi Zhang**, Lixin Liu, Yuxiao Dong, Cuiping Li, Hong Chen and Hongzhi Yin. Open-World Semi-Supervised Learning on Graph. In *ICDE'24*, pages 2723–2736, May. 2024
- [6] **Lingxi Zhang**, Jing Zhang, Yanling Wang, Shuling Cao, Xinmei Huang, Cuiping Li, Hong Chen and Juanzi Li. FC-KBQA : A Fine-to-Coarse Composition Framework for Knowledge Base Question Answering. In *ACL'23*, pages 1002–1017, Jul. 2023
- [7] **Lingxi Zhang**, Xirui Ke, Haoyang Li, Xingmei Huang, Zhonghui Shao, Shulin Cao and Xin Lv. A Survey on Complex Factual Question Answering. In *AI Open'23 (Vol 4)*, pages 1–12, 2022
- [8] Jing Zhang, Bo Chen, **Lingxi Zhang**, Xirui Ke and Haipeng Ding. Neural-symbolic Reasoning on Knowledge Graphs. In *AI Open'21 (Vol 2)*, Pages 14–35, 2021

RESEARCH EXPERIENCE

Chili Lab, Rice University

Houston, TX

Ph.D. student, supervised by Prof. Hanjie Chen

Multi-Agent System Safety

09 / 2025 – Present

- **Project (lead): Rethinking Safety in LLM-Based Multi-Agent Systems.**

– Introduced three near-benign MAS attacks (Slow Drift / Benign Wrapper / Chaos Seeding) that make malicious messages embedding-close to benign ones and bypass embedding-based defenses. Proposed a confidence-guided defense using token-level uncertainty to prune/down-weight messages, improving robustness across models and tasks.

DATA Lab, Rice University

Houston, TX

Ph.D. student, supervised by Prof. Xia Hu.

LLM-Based Multi-Agent Systems

09 / 2024 – 08 / 2025

- **Project (lead): Decoupled Multi-Agent Framework for Text Style Transfer.**
– Developed an automatically decoupled multi-agent framework for complex entangled style transfer. Incorporated a self-check strategy for iterative refinement, achieving higher average scores on CDS (style +2.8%, content +4.4%).
- **Project (lead): Multi-Agent Efficiency Benchmark.**
– Benchmarked recent multi-agent frameworks and topologies with a developed metric that measures per-message ROI, rewarding agents that deliver valuable information concisely and cost-effectively.

Georgia Institute of Technology

Atlanta, GA

Research intern, supervised by Prof. Chao Zhang.

Hallucination Issue in Large Language Model

04 / 2023 – 06 / 2024

- **Project (lead): Self-Guided Retrieval Augmentation for Black-Box Large Language Models.**
– Enhanced a black-box LLM for QA with an adapted retriever trained on LLM-annotated data, improving F1 by 8.3% on NQ and 7.4% on MMLU while reducing hallucination and enhancing few-shot transferability.

Knowledge Engineering Group, Renmin University of China

Beijing, China

Graduate research assistant, supervised by Prof. Jing Zhang.

Generalization Issue in Knowledge Base Question Answering (KBQA)

05 / 2022 – 07 / 2023

- **Project (lead): A Fine-to-Coarse Composition Framework for Knowledge Base Question Answering**
– Decoupled the KB query language into fine-grained components to reduce entanglement and overfitting. The fine-to-coarse framework improves generalization, achieving SOTA on GrailQA and WebQSP while running 4× faster.
- **Project (lead): A LM-Enhanced Solution for Question Answering over Multiple Knowledge Bases.**
– Proposed a learn-then-reason KBQA approach integrating knowledge into LLM for better generalization and end-to-end reasoning. KBLLaMA outperforms baselines and GPT-4, gaining 3.8% on GrailQA and 9.8% on Bio-chemical.

Open-World Semi-Supervised Learning (SSL) on Graph

01 / 2023 – 06 / 2023

- **Project (participant): Alleviating the imbalance issue to develop open-world SSL on graph.**
– Proved that the imbalance of intra-class variances is a key challenge for graph SSL.

Reasoning and Question Answering over Knowledge Base

04 / 2021 – 06 / 2022

- **Project (lead): Surveying approaches on knowledge graph/base and group them in an unified framework.**
– Surveyed all recent complex factual question answering models across various data sources, list the similarities among these approaches, and group them into the analysis-extend-reason framework.

WORK EXPERIENCE

Applied Scientist Intern, Microsoft

Redmond, WA

- **Self-evolving skill optimization through actor-critic.**

05 / 2026 – 07 / 2026

– Designed a RL framework to self-evolve skill optimization for Word agents.

Research Intern, ByteDance

Beijing, China

- **Document question answering (QA) system about cars.**

06 / 2021 – 06 / 2022

– Collected QA pairs from the web and designed a ranking loss to fine-tune the BERT model with multiple documents.
– Launched online, which increased the recall rate by 220% and the click through rate (CTR) by 1.3%.

Research Intern, Zhipur

Beijing, China

- **Knowledge-ground dialog system based on pre-train language model (PLM).**

07 / 2019 - 02 / 2020

– Led the knowledge probing group which focus on mining the knowledge fact that stored in PLM's in-parameter.

AWARDS & HONORS

- Ken Kennedy Institute Computational Science and Engineering Graduate Recruiting **Fellowship** 08 / 2024
- Bronze Medal, **Asia-East Continent Final, ACM-ICPC** (Top 30%) 12 / 2020
- National First Prize, **Contemporary Undergraduate Mathematical Contest in Modeling** (Top 0.7%) 09 / 2020
- Best All-Girl Team, **Nanjing Regional Contest, ACM-ICPC** (Top 1) 10 / 2019